Remote Particle Trajectory Tracking using Event-Based Vision Streams

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Abstract

Classical particle tracking using frame-based Active Pixel Sensor (APS) struggles with poor lighting conditions, scenes with high dynamic range, dynamic changes in illuminance and restricted frame rate. We extend Particle Tracking Velocimetry (PTV) to such difficult environmental conditions by utilizing Event-Based Vision (EBV) sensors, which asynchronously generate timestamped events in response to pixel intensity changes.

We present a remote PTV system that uses stereo 2D event-based camera outputs to reconstruct and stream 3D particle events over the network for remote particle trajectory tracking in real time. We quantify the accuracy of the 3D point reconstruction and the trajectories against the ground truth obtained from an infrared motion capture system. Further, we decompose the end-to-end latency into the processing latency (from event capturing to triangulation) and the remote network transport and tracking latency and show how to feedback the transport layer congestion signal to filter and transmit only the less noisy events. Our results quantify the impact of different particle counts on the accuracy and latency. We observe that uniform random subsampling of event streams accelerates processing and tracking without accuracy impairment; however, the "burstiness" of the event streams over the network may lead to particle misclassification and tracking resets.

CCS Concepts

• **Computing methodologies** → **Image processing**.

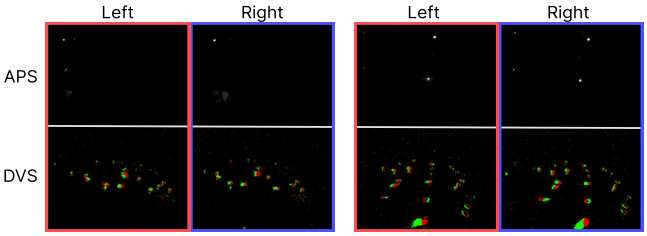
Keywords

Stereo Event-Based Vision, Streaming-based Particle Tracking

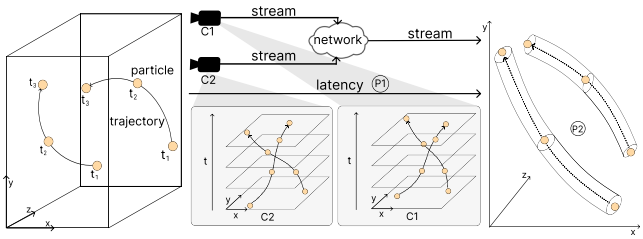
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1 Introduction

Particle tracking is a core task in fluid and gas motion analysis in the field of fluid mechanics [1, 7, 28, 30], insect tracking [2], drone detection and navigation [5]. The term *particle* typically refers to an object in a medium whose passive motion (e.g., algae in a fluid) reflects the flow dynamics in the system [28], or whose active motion (e.g., an insect or a drone moving in the air) is of interest. Our particle sizes of 0.8 and 2 cm correspond to the body size of small animals or insects, suggesting that this particle tracking setup can be extended for tracking animal swarms [2, 21]. *Remote* particle tracking is relevant in scenarios where on-device evaluation is impractical, e.g., due to limited resources or harsh environments.



(a) Synchronous output frames from Active Pixel Sensors (APS) and Dynamic Vision Sensors (DVS) of a stereo DAVIS346 camera setup capturing a particle movement scene in challenging illuminance conditions (1.5 lx).



(b) Sketch of our remote 3D particle tracking system.

Figure 1: (a) Particle tracking using the APS sensors fails as the obtained frames are almost entirely dark. As the DVS output captures the normalized temporal changes in illuminance particle tracking is viable. (b) Our setup is based on two event cameras that produce 2D-temporal pixel event streams which are 3D-triangulated and streamed over the network to a particle tracking algorithm. Challenge P_1 concerns the end-to-end latency, i.e. how tracking lags behind capturing, while challenge P_2 concerns the accuracy of the DVS output-based particle tracking.

Challenges in particle tracking include (i) accurately identifying individual particle trajectories, (ii) time-dynamic illumination conditions, and (iii) required high data rates for remote frame-based tracking. Fig. 1a shows how traditional frame-based approaches using the Active Pixel Sensor (APS) struggle under challenging illuminance conditions. As evident from the APS row in the figure, tracking using an image frame approach fails in this example due to particle misdetection.

Event cameras capture scene information as a time-asynchronous stream of pixel samples triggered by changes in individual

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pixel luminance. Fig. 1a shows corresponding synchronized scenes as captured by the Dynamic Vision Sensor (DVS). The DVS is ideal to capture motion as it triggers an event per pixel when the light intensity changes by a relative threshold since the previous event. This leads to the bi-colored events in the DVS row of the figure for positive and negative threshold crossing, respectively. Compared to frame-based approaches, event-based vision offers high temporal resolution, high dynamic range, and reduced motion blur [11, 17, 29] leading to a strong potential for particle tracking applications. However, the existing methods [34–36] are limited to pulsed illumination and single Event-Based Vision (EBV) sensor use.

In this work, we approach the challenge of remote particle tracking using the system depicted in Fig. 1b. This is the first work to tackle remote particle tracking using stereo event-based cameras, where we investigate factors relating the use of dynamic vision sensors to the tracking accuracy and the end-to-end latency.

We formulate the accuracy and remote tracking latency as optimization problems. We present the system approach to remote particle tracking seen in Fig. 1b. It consists of (i) capturing and 3D reconstruction from stereo event-based vision sensor streams, (ii) aggregated event data streaming over a communication network, and (iii) a remote Particle Tracking Velocimetry (PTV) module.

Our experiments show that the accuracy of the reconstructed particle positions closely matches the ground truth provided by a motion capture system (OptiTrack [24]) and that the particle processing and the tracking latency increases dramatically with the number of particles. In particular, the accuracy evaluation shows high coverage of the raw points and low deviation (Chamfer distance) (on average 2.3 cm). The latency evaluation shows that over 75% of the 3D points are sent and produce an estimate output on the receiver side within ~300 ms, but also that delay spikes occur occasionally. We find that uniform random subsampling of events accelerates processing and tracking without reducing accuracy and that the burstiness of the event streams may lead to particle misclassification and track resets. Finally, the sender feeds a congestion signal obtained from the congestion control algorithm back to an application layer event filtering mechanism to guide the event subsampling. Essentially, the filter ranks the events by quality, i.e. using a noise estimate, and delivers the less noisy events to the transport layer upon congestion.

2 Related Work

2.1 Frame-Based Imaging for PTV

In frame-based particle tracking, the particles are captured at fixed frame intervals and each particle is only updated when a new frame is captured. Particles are detected by thresholding, i.e., only keeping pixels that are brighter than a certain threshold, clustering and linking those clusters across frames (e.g., using nearest neighbor distances). The work in [7] gives an overview of PTV methods, including particle identification algorithms, algorithms to estimate the particle location in 2D and 3D space, and algorithms to track the particles in successive images.

Conventional frame-based approaches using active pixel sensors have been classically applied in the area of PTV. It is known, however, that the application of those sensors to PTV is limited by motion blur (cf. Fig. 2) and a constrained dynamic range [11–13].



Figure 2: APS vs. DVS outputs for a fast moving particle recorded by the DAVIS346. The grayscale APS frame (left) shows visible motion blur, while the DVS events (right) show the trajectory (green) and the particle without blur (red).

Fig. 1a shows examples of frame outputs from active pixel sensors in challenging illumination conditions to demonstrate the difficulty of deploying classical frame-based imaging for PTV in such conditions. Additionally, frame-based imaging generates redundant data, because each frame captures the full scene.

Frame-based imaging systems for particle tracking in fluids and gases are further described in [1, 28, 38]. The authors of [38] present a frame-based approach to PTV. In comparison to [1], which only considers 2D representations, the authors of [38] focus on the 3D representation of the trajectories. The 3D reconstruction is achieved through classical virtual stereo vision, where one camera observes the reflection of two differently angled mirrors.

2.2 Event-Based Vision Sensors

Event-based vision sensors differ fundamentally from conventional active pixel sensors known from frame-based cameras [12, 13, 29]. Instead of capturing full images at fixed inter-frame intervals, e.g. 30 frames per second, in event-based vision each pixel circuit operates independently and asynchronously. The first lab proof-of-concept DVS was released in 2008 with 128 ×128 px and 120 dB dynamic range [20]. The DVS sensors of the DAVIS346 event-cameras used in this work possess a resolution of 346 ×260 px and 120 dB dynamic range [17] with an integrated and synchronized additional APS sensor that captures 346 ×260 px grayscale frames at 40 fps corresponding to ~29 Mbps.

DVS produce events $(x_i, y_i, t_i, p_i)_{i\in\mathbb{N}}$ at the pixel coordinates (x_i, y_i) at time t_i with polarity $p_i \in \{+1, -1\}$ when the relative changes in the log luminance $\Delta L(x_i, y_i, t_i)$ defined as $L(x_i, y_i, t_i) - L(x_i, y_i, t_{i-1})$ cross the threshold $p_i \cdot \theta$ with free parameter $\theta > 0$ [11]. Here, the luminance is measured by the log photocurrent I as $L = \log(I)$. In a nutshell, when $|\Delta L(x_i, y_i, t_i)| > \theta_i$ an event fires at the pixel coordinates (x_i, y_i) with a polarity corresponding to sign of $\Delta L(x_i, y_i, t_i)$. Hence, the DVS pixels capture changes in the logarithmic intensity of the luminance when a certain threshold is exceeded. The output of an event-based camera is thus a stream of asynchronous events, containing a timestamp in microseconds, x- and y-coordinates of the pixel location and the binary polarity. DVS are known for a reduced motion blur, especially in low illumination conditions, in comparison to APS, as discussed in [15].

Event-based vision sensors such as DVS [13] offer various advantages over frame-based approaches, including high temporal resolution, high dynamic range and reduced motion blur [12, 13, 29]. Additionally, frame-based approaches produce a fixed data-rate, not considering the scene's activity (~29 Mbps for the DAVIS345 APS at 40 fps) while event-based approaches can generate substantially

less data. In our setup we measured a maximum of 60k events/s per camera, leading to a data rate of ~3.8 Mbps.

An overview of state-of-the-art works, datasets, devices and tools related to event-based vision are given in [29] and a detailed overview of the underpinnings and some applications of event-based vision is in [11]. Beyond the mentioned advantages of event-based vision, some studies analyze the potential and limitations of these sensors in specific use-cases: The authors of [31] compare frame-based and event-based cameras for fast moving objects, e.g. flapping-wing robots, showing the advantages of event-cameras in terms of power efficiency. The authors of [12] postulate that lower-resolution event-based vision cameras can outperform high-resolution event-based vision cameras especially in low illumination and for fast moving objects due to lower per-pixel temporal noise. Additionally, the authors of [13] detail the internal operation of the DVS and provide camera configurations considering different application requirements and states of the scene.

2.3 Event-Based Imaging for PTV

Event-based DVS inherently provide advantages such as High Dynamic Range (HDR) and reduced motion blur [27], offering a promising solution for tracking fast moving objects such as particles. While frame-based approaches of particle tracking with diverse setups are established, we find that only a few works explore the potential of event-based vision for tracking particles [34–36]. The authors of [34] show the feasibility of monocular event-based vision for Particle Image Velocimetry (PIV) in the context of fluid flow measurements, where 2-dimensional velocity fields are obtained. In contrast, we investigate stereo event-based vision with two event cameras and 3D reconstruction of the tracked particles and their trajectory representation and tracking.

The authors of [35] introduce pulsed illumination to quantify the latency of the system for exact velocity calculation for Event-Based Imaging Velocimetry (EBIV) with one event-based sensor. Further, the work in [36] presents PTV with event-based vision in the analysis of a turbulent boundary layer, where a layer of fluid is influenced through a close by surface. Despite the advantages such as high temporal resolution, high dynamic range and reduced motion blur, event-based vision and 3D reconstruction for particle tracking remain largely unexplored.

3 Problem Statement

Next, we introduce the formal problem statement behind studying the tracking accuracy and the end-to-end latency of remote 3D reconstruction and tracking of particles using event-based vision.

Accuracy: We seek to estimate a set of reconstructed 3D point positions $R = \{r_i\}_{i\in\mathbb{N}}$ with $r_i \in \mathbb{R}^3$ from the stereo event streams $S^1(t_i), S^2(t_i)$ such that the reconstructed points R closely match the set of the true particle points $O = \{o_i\}_{i\in\mathbb{N}} \in \mathbb{R}^3$ with $o_i \in \mathbb{R}^3$ that are obtained from a reference ground truth motion capture system. An event stream $S^j(t_i) = (x_i^j, y_i^j, t_i)_{t_i\in\mathbb{R}}$ denotes the x- and y-coordinates firing an event at sensor $j \in \{1, 2\}$ at times t_i. Note that finite accuracy timestamping may yield multiple events $e^j(t_i)$ that possess the same timestamp value t_i. The goal of the 3D reconstruction is to map $e^1(t_i) \in S^1(t_i)$ to $e^2(t_k) \in S^2(t_k)$. Evidently, the event positions in $S^1(t_i), S^2(t_k)$ vary and the timestamping on the sensors may differ too. The mapped events (e^1, e^2) allow a direct 3D reconstruction based on triangulation to produce the set of estimated 3D points R. We seek an event mapping that minimizes the objective function $\mathcal{L}_{\text{acc}}$ as in

$$\min_{\chi} \ \mathcal{L}_{\text{acc}}(R, O) \tag{1}$$

where $\mathcal{L}_{\text{acc}}$ is the difference metric between the reconstructed and the ground truth point sets R and O, respectively. The vector χ contains the configuration parameters of the reconstruction, i.e., the mapping between the events (e^1, e^2), e.g. the maximum temporal difference between two mapped events $\max_{i,k} |t_k - t_i|$, the maximum vertical pixel offset $\max_{i,k} |y_k - y_i|$, or the maximum pixel disparity. The difference metric $\mathcal{L}_{\text{acc}}$ is given in this work, e.g. by the coverage Cov, precision Prec, Euclidean norm $||\cdot||_2$ based nearest neighbor distances d_{NN} and the Chamfer distance d_C.

Latency: We seek to minimize the latency between the sender-side particle motion and the receiver-side trajectory estimation. This end-to-end latency is decomposed using three timestamps: The timestamp of the ith captured event at the event-based sensor denoted t_i, the time point at which a 3D point is constructed that is denoted $t_{p,i}$ and, finally, the time point at which the particle tracking algorithm at the receiver outputs a 3D point as part of a trajectory that is denoted $t_{s,i}$. Note that the number of raw captured events $|\{t_i\}|$ is greater or equal to the number of reconstructed 3D points $|R|$, where $|\cdot|$ denotes the set cardinality. Hence, the sought time differences are only computed with respect to the reconstructed 3D points. The latencies for processing and remote tracking of a reconstructed point triangulated from the ith event are respectively as $L_{\text{proc},i} = t_{p,i} - t_i$ and $L_{\text{track},i} = t_{s,i} - t_{p,i}$. The end-to-end latency is then obtained as $L_{\text{lat},i} = t_{s,i} - t_i$. This end-to-end latency $L_{\text{lat},i}$ is a function of the number of raw events $|\{t_i\}|$, the number of reconstructed points $|R|$ as well as streaming parameters such as the number of points that are packed into one data packet.

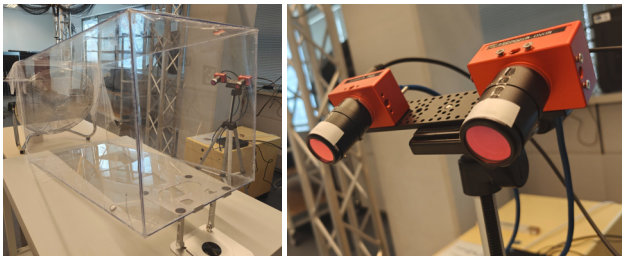


Figure 3: (Left) The experimental setup incl. the particle box, reflective particles and a fan for inducing random particle movement. The stereo camera setup is seen in the back. (Right) Two DAVIS346 cameras [17] with custom infrared filter lenses on a stereo mount (14 cm apart).

4 Event-based particle tracking

We present our system for event-based stereo particle tracking with 3D point reconstruction and real-time streaming of the triangulated points to a remote tracking algorithm. The 3D points of the tracked particles are presented as trajectories and validated against the ground truth data obtained from an OptiTrack motion capture system [24]. Further, we study the latencies of processing and streaming the triangulated points to the tracking receiver.

The system consists of (i) a scene with retro-reflective particles, (ii) a stereo camera setup with two DAVIS 346 event-cameras capturing events of the scene, (iii) a processing unit[1] that performs calibration and triangulation to then output and send 3D points, (iv) a receiver with a MATLAB-based tracking algorithm that processes the triangulated points, and (v) an OptiTrack motion capture system for the ground truth reference. The OptiTrack motion capture system has 8 OptiTrack Prime 13 Motion Capture Cameras [26] (1280 ×1024 px, field of 56° ×46°) installed in an approximately 6.5 m × 4.5 m empty lab area at a height of ∼2 m and the ground truth data for the evaluation is extracted via `Motive` [25].

Fig. 3 shows a custom-built box with size 0.5 m × 0.5 m × 1 m for the experiments containing styrofoam particles with a retro-reflective surface and is angled at 10°. We use homogeneous particles of a diameter of 2 cm or 0.8 cm. An external fan induces random particle motion. The particle movement is captured simultaneously by the event cameras and the motion capture system.

The stereo setup consists of two parallel DAVIS 346 cameras [17] mounted in a fixed configuration in the same height with a distance of 14 cm as depicted in Fig. 3. The cameras have a distance of ∼40 cm to the particle box, meaning the distance of the particles to the camera lenses has a range of 40 cm to 100 cm. Each DAVIS346 camera combines a DVS sensor (346 ×260 px, 120 dB dynamic range) with a conventional APS (346 ×260 px, grayscale, 56.7 dB dynamic range). The measured illuminance during the standard illumination experiments averages to ∼ 300 lx and for the low illumination experiment to ∼ 1.5 lx as measured by a digital lux meter. Measurements are taken at multiple locations inside and outside the particle box within the camera scene under standard laboratory conditions (fluorescent ceiling lights on, blinds closed). Since the motion capture system uses infrared light for tracking, we equip the cameras with infrared lens filters to reduce interference.

We use the `dv-gui` framework to process the 2D-temporal event streams [18]. We calibrate the cameras using the calibration modules in the framework [16]. Hence, we obtain calibration matrices that are used to compute the projection matrices [14]. For the stereo event matching of the left and right camera, we suppress the polarity p_i of the events (x_i, y_i, t_i, p_i), defined in Sect. 2. By matching events from both cameras regardless of polarity we reduce the impact of noise as noisy pixels are less probable to appear simultaneously in both cameras. The timestamps of both cameras are synchronized prior to the experiments and the events are filtered with a DVS noise filter [19] before triangulation. This filter combines a refractory period (minimum interval between same pixel events) and hot-pixel suppression (pixels that are always *on*) and is hence suitable for static camera setups.

With respect to the accuracy problem stated in (1) we perform a grid search and a visual analysis, leading to the optimized mapping of camera events with the following configuration: The matching candidates are selected with the boundaries $t_{\max} = 150\,\mu$s as the temporal window for candidate matching, $y_{\text{tol}} = 15$ px as the maximum y-coordinate offset and $d_{\min} = 5$ px and $d_{\max} = 150$ px as the disparity bounds. Two matching candidates k and i are ranked by a weighted score $\in [0, 1]$ with the weights $\alpha = 0.5, \beta = 0.4$ and $\gamma = 0.1$ over the normalized difference in timestamp $|t_k - t_i|$, y-coordinate-offset $|y_k - y_i|$ and disparity deviation as in:

$$s_{k,i} = \alpha \frac{|t_k - t_i|}{t_{\max}} + \beta \frac{|y_k - y_i|}{y_{\text{tol}}} + \gamma \frac{\left||d_k - d_i| - \frac{1}{2}(d_{\max} + d_{\min})\right|}{\frac{1}{2}(d_{\max} - d_{\min})} \quad (2)$$

An event pair is selected based on this ranking, considering the minimization of the score. The thresholds are selected to account for streaming latency, calibration inaccuracies and particle motion while remaining small enough to maintain accuracy. The selected match is triangulated with a reconstruction module to perform real-time 3D reconstruction using standard triangulation in OpenCV [23].

The triangulated 3D-points are either transmitted via the QUIC transport protocol or via plain UDP to enable low latency and fast parsing. We postulate that missing events do not harm the tracking accuracy as shown in Sect. 6. We investigate event subsampling and congestion-based filtering in Sect. 6.

Finally, the streamed packets are received and parsed at the receiver using MATLAB for trajectory computation and representation. To spatially group multiple received and triangulated points that represent the same particle at the same time we use the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm [8]. DBSCAN groups nearby points into clusters based on a maximum distance between points (∼1 cm) and a minimum number of points per cluster (2 points). Single points or outliers are treated as noise and discarded. The center of each cluster is then treated as a single detection and passed to MATLAB's Global Nearest Neighbor (GNN) `trackerGNN` [3, 22], enabling multi-object tracking using three steps: prediction, assignment and confirmation/deletion. During prediction, the future position of the particle is estimated using Kalman filtering. In the assignment step, the incoming event detections are associated to the existing tracks considering the GNN of the predicted and the measured event positions. The following confirmation and deletion rules are applied given a timestep length of 20 ms for the tracker updates. The confirmation rule uses a `ConfirmationThreshold` = (1, 3), meaning that a new track is confirmed if it receives at least one detection in the last three tracker update steps. A track is deleted if no detection was matched to that track in five of the last twelve update steps (`DeletionThreshold` = (5, 12)).

5 Experiments

We evaluate the remote particle tracking system in two different scenarios. First, we focus on accuracy measurements of the raw and 3D reconstructed points as well as the trajectory points. We compare with the ground truth from the motion capture system. Second, we focus on streaming measurements examining the end-to-end latency when considering the streaming setup, the processing time of the 3D reconstruction module, the stream throughput, and network congestion-based application layer event filtering. Unless otherwise noted, the experiments use fixed particle counts $\{1, 2, 3, 5, 10\}$, referring to the total number of particles in the box. Each experiment is repeated independently 25× for a 10 sec duration each.

The network setup is depicted in Fig. 4. We use a dedicated switched network with 1Gbps links without cross traffic. In our measurements we do not observe any additional queueing-based or interference-based delays, jitter or packet loss except for the

[1]Experiments run on Lenovo ThinkCentre M90q Gen 3 with Ubuntu 22.04.5 LTS, 12th Gen Intel Core i5-12500, 8 GB RAM.

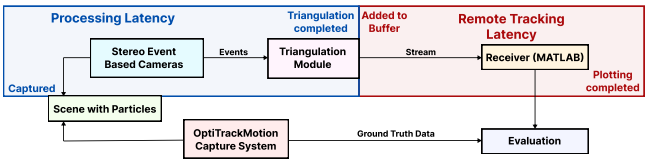


Figure 4: Overview of the networked tracking system showing the streaming latency components: The processing latency is in blue (left) and the remote tracking latency is in red (right). The packet data stream between the triangulation module and the receiver is transmitted over a dedicated switched network.

experiments on congestion-based application layer event filtering where we synthetically induce a variable bottleneck using `tc`.

5.1 Accuracy Measurements

The purpose of this experiment is the evaluation of the accuracy of the trajectories based on the triangulated points against the ground truth retrieved by the motion capture system. This allows determining if our event-based stereo particle tracking system can provide reliable trajectories.

We calibrate the experiments using a single particle for lowest complexity and to allow for simple evaluation of the reconstruction accuracy before increasing the number of particles. A quantitative evaluation of the accuracy of the trajectories based on (1) is presented in Section 6.1. In addition, for a qualitative analysis, we visually compare the reconstructed 3D trajectories to the ground truth as shown in the one particle example overlay in Fig. 5. The red trajectory represents the ground truth, while the blue toned trajectory parts are retrieved from the reconstructed points. If for any given particle no events arrive for some time, i.e., the particle is not visible or due to noise, the tracker misclassifies the events and resets the trajectory (hence the new blue tone). We execute the experiments with different particle sizes and different distances between the sensors and the particles. The evaluation of the variation of the particle size is sensible for use cases such as tracking algae flakes in a fluid as these appear in differing sizes. The system's adaptability to varying distances of the objects to the sensors is important for deployment in different domains where the physical setup might restrict the distance of the cameras to the target.

5.2 Streaming Measurements

These experiments aim to inspect the streaming properties of the EBV streams. For example we examine if the low latency advantage of event-based vision sensors is preserved when applying the 3D reconstruction and streaming pipeline. As depicted in Fig. 4 the system introduces latencies within the matching and triangulation operations as well as network streaming and trajectory tracking delays. To this end, we measure the processing latency $L_{\text{proc},i}$ and remote tracking latency $L_{\text{track},i}$ independently and report the results. Further, we capture the transmitted data streams to assess the throughput and burstiness of the streams between the sensor-side triangulation and the tracking algorithms at the remote receiver. Finally, we capture the congestion feedback provided by the QUIC transport protocol to signal to the application to filter out events that are more corrupted by noise before transmission.

Event Processing Latency: The processing latency $L_{\text{proc},i}$ as defined in Sect. 3 and depicted in blue in Fig. 4 describes the delay time between capturing the ith event, which is represented by the event's timestamp and adding the triangulated event into a transmission buffer for transmission.

Remote Tracking Latency: The remote tracking latency $L_{\text{track},i}$ as defined in Sect. 3 and depicted in red in Fig. 4 gives the time between adding the event into the transmission buffer for sending $t_{p,i}$ and the time of the trajectory output $t_{p,i}$. Components of this latency are mainly the processing time in the trackerGNN as well as a waiting time of max. 2 ms in the transmission buffer.

6 Evaluation

6.1 Accuracy Evaluation

Next, we describe the evaluation metrics and the accuracy results.

Evaluation Metrics: We quantify the accuracy using point-set distance metrics between the 3D reconstructed points from the stereo event-based vision cameras and the ground truth from the OptiTrack motion capture system.

Recall that O is the set of ground truth points and R is the set of 3D reconstructed points from the event-based camera streams. For any points x, q and point set Q, let $\|x - q\|$ denote the Euclidean distance and define the nearest neighbor distance as $NN(x, Q) = \min_{q \in Q} \|x - q\|$. The accuracy metrics are then defined as follows.

Coverage $\text{Cov}(\rho)$: Fraction of ground truth points that have at least one reconstructed point within radius ρ.

$$\text{Cov}(\rho) = \frac{1}{|O|} \left| \{ o \in O \mid \min_{r \in R} \|o - r\| \leq \rho \} \right|. \tag{3}$$

Precision $\text{Prec}(\rho)$: Fraction of reconstructed points that are closer than ρ from any ground truth point.

$$\text{Prec}(\rho) = 1 - \frac{1}{|R|} \left| \{ r \in R \mid \min_{o \in O} \|r - o\| > \rho \} \right|. \tag{4}$$

Average Nearest Neighbor Distance: The average nearest neighbor distance in one direction for either point set O or R are

$$d_{NN}(O, R) = \frac{1}{|O|} \sum_{o \in O} NN(o, R), \tag{5}$$

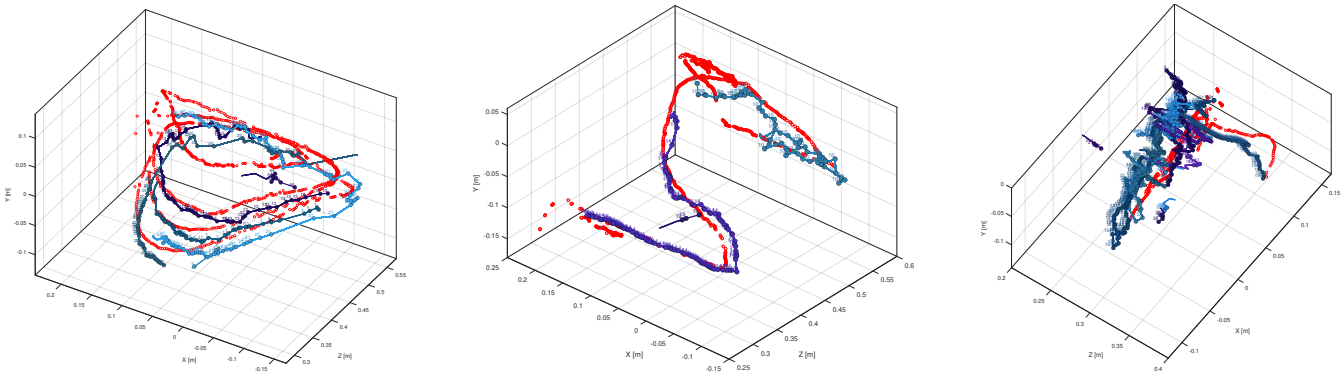
$$d_{NN}(R, O) = \frac{1}{|R|} \sum_{r \in R} NN(r, O). \tag{6}$$

Chamfer Distance $d_C(O, R)$: For both point sets O and R, the average of the distances from each point $o \in O$ to its nearest neighbor in R and from each point $r \in R$ to its nearest neighbor in O [37] is taken as

$$d_C(O, R) = \frac{1}{2} d_{\text{NN}}(O, R) + \frac{1}{2} d_{\text{NN}}(R, O). \tag{7}$$

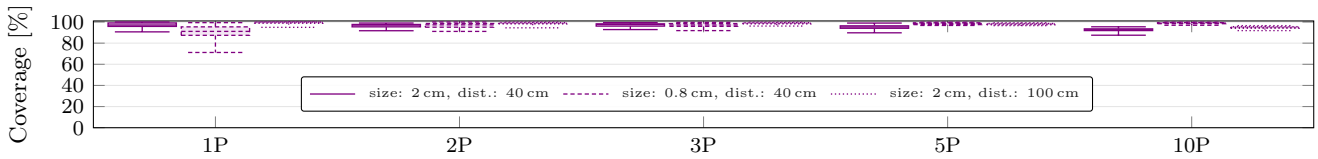
Results: Next, we compare the ground truth from the motion capture system to the raw 3D points from the event cameras as well as the tracked trajectories.

Raw Points. Figs. 6a-6c show the results of the evaluation metrics for experiments with different particle numbers, sizes and distances between the sensors and the particles. For every fixed particle number, the left boxplot represent the experiment with 2 cm particles at 40 cm distance, the middle boxplots those with 0.8 cm particles at 40 cm distance and the right boxplots those with 2 cm particles at 100 cm distance. The metrics compare the ground truth that is retrieved by the OptiTrack motion capture system against the raw

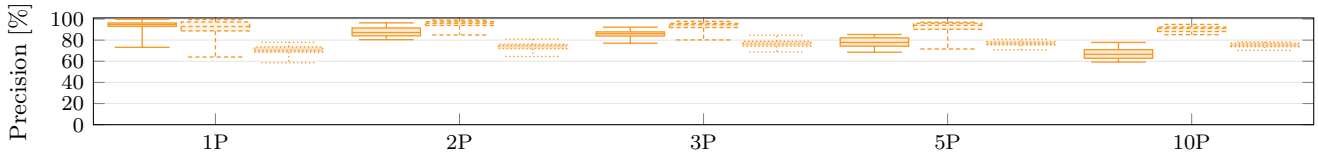


(a) Normal illumination. (b) Normal illumination. (c) Low illumination.

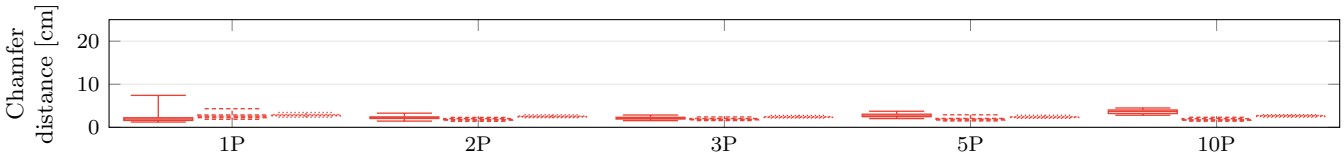
Figure 5: Qualitative results for 10 s experiments with one particle. The ground truth is shown in red, while the blue tones show the estimated trajectories. The blue tone changes when the tracker resets such as when events are missing or the new events do not match the Kalman bounds for the current trajectory. The tracked trajectories closely align with the ground truth, although the low illumination (cf. Fig. 1a) of (c) yields more frequent tracker resets.



(a) Coverage of the raw points: particle size 2 cm, distance 40 cm; particle size 0.8 cm, distance 40 cm; particle size 2 cm, distance 100 cm.



(b) Precision of the raw points: particle size 2 cm, distance 40 cm; particle size 0.8 cm, distance 40 cm; particle size 2 cm, distance 100 cm.



(c) Chamfer distance of the raw points: particle size particle size 2 cm, distance 40 cm; particle size 0.8 cm, distance 40 cm; particle size 2 cm, distance 100 cm.

Figure 6: Accuracy evaluation with respect to the raw points for different particle counts $\{1, 2, 3, 5, 10\}$P. Comparison of coverage, precision and Chamfer distance over particle counts for raw point clouds, for 2 cm and 8 mm particle size and 40 cm and 100 cm distance between particle box and sensors. The left boxplots represent the experiment with 2 cm particles at 40 cm distance, the middle boxplots those with 0.8 cm particles at 40 cm distance and the right boxplots those with 2 cm particles at 100 cm distance.

reconstructed points after alignment with the ground truth trajectories. Those raw points are then used by the MATLAB trackerGNN to compute the trajectories. We evaluate the coverage and precision with a radius of $\rho = 5$ cm. We consider this radius to strike a balance between event mapping accuracy and the impact of noise.

The coverage as depicted in Fig. 6a remains consistently high with values above 70% across all experiments and an average of 96%, showing that most ground truth points have a reconstructed neighbor from the event-based camera within the given radius ρ. We observe that the precision $\text{Prec}(\rho)$ has an average of 83%, which is depicted in Fig. 6b with a decreasing trend when increasing the number of particles. We attribute the combination of a high coverage and varying precision in some of the experiments to noise in the reconstructed point data which increases with the number of particles and can be observed by visual inspection.

The Chamfer distance $d_C(O, P)$ averages the distances from each point in each point set to its nearest neighbor in the other point set. The Chamfer distance in Fig. 6c is below 7.5 cm in all the experiments. Notably the average of the Chamfer distance is 2.3 cm across all of the 375 experiments, which is roughly the diameter of one particle.

Trajectory Points. Figures 7a-7c show the results of the evaluation metrics comparing the ground truth against the tracked trajectory points by the trackerGNN algorithm after alignment of the two point clouds. An overlay of the ground truth data and the trajectories can be seen in Fig. 5.

We observe in Fig. 7a coverage $\mathrm{Cov}(\rho)$ with an average value of 85%. This shows that most ground truth points have a reconstructed neighbor from the event-based camera within a radius ρ. The precision $\mathrm{Prec}(\rho)$ shows an average of 55% with a decreasing trend when increasing the number of particles and especially low values for the experiment with a larger distance to the sensors. We find that the `trackerGNN` trajectory tracking algorithm is not recognizing all of the raw points as correct trajectory points, explaining the decrease from the raw points to the trajectory points and that the matching parameters are not optimized for the distance variation.

The results depicted in Fig. 7c show that the mean Chamfer distance of all experiments is 7.6 cm with a slightly increasing distance in the number of particles. Overall, this shows a good tracking performance with an increase in noise when increasing the number of particles.

High dynamic range experiment: Next, we describe the low illumination tracking experiment from Sect. 1 with frame extractions given in Fig. 1a. For that case study, we conduct the experiment with one particle where Fig. 5c shows the ground truth from the OptiTrack motion capture system in red and the estimated particle trajectories in blue tones. While the active pixel sensors fail to capture enough data for tracking in such low-illumination setting, the event-based vision sensors capture the events shown in the figure. We observe a good match between the event-based tracking and the ground truth albeit noise increase.

6.2 Latency Evaluation

We evaluate the latency of the system consisting of the two sections depicted in Fig. 4, i.e., the event processing latency and the remote tracking latency which includes the network transmission.

Event Processing Latency Evaluation: For the event processing latency we evaluate the delay between capturing the events and the triangulation output.Recall that the processing latency $L_{\mathrm{proc},i}$ describes the time between capturing the ith event and handing that event to the transport layer buffer (of maximum waiting time of 2 ms) for transmission.

The processing latency results in Fig. 8a-8b show a sharp increasing trend with higher particle count. While the processing latency for 1 and 2 particles remains low with medians below 3 ms, the processing latency increases by 3 orders of magnitude for 10 particles. We observe that the bottleneck is the event matching loop iterating over a much larger number of events (including noise) to find the best match for the triangulation. The increase in the event count per second can be seen in Fig. 9. We observe after applying DVS noise filtering that there are on average $5 \cdot 10^3$ events per second per camera for 1 particle while there are $27 \cdot 10^3$ events per second per camera in the stream for 10 particles. We observe that the increase in processing latency for higher particle count is caused by the iterations of the processing loop and not inherent to the event-based vision sensor.

Remote tracking Latency Evaluation: We evaluate the remote tracking latency L_{track} from Sect. 3 by measuring the time from the events being added to the transport layer buffer on the sender side to the time point of the trajectory output.

Evaluation Metrics: The remote tracking latency L_{track} is computed across hosts with synchronized clocks using Network Time Protocol (NTP) [6] with an average offset of 0.4 ms. The sender batches 30 reconstructed 3D points per packet and flushes the buffer after at most 2 ms or when 30 events are collected. This bounds the buffering time on the sender to at most 2 ms.

Results: Figure 8c shows the CDF of the remote tracking latency. We observe that the median of the remote tracking latency remains between $110 - 170$ ms for all particle counts. This suggests that the streaming and tracking capacity is not saturated. Further, as seen in Fig. 9, the number of captured events per second per particle count differs significantly. The number of reconstructed 3D points is on average at 10^2 or 10^3 triangulated points per second for one particle or 10 particles, respectively. Figure 8c further shows that all particle counts follow a similar remote tracking latency distribution. However, the variance of this distribution is large which is mainly attributed to the tracking algorithm.

6.3 Throughput Evaluation

Evaluation Metrics: We consider remote tracking experiments with different number of particles within the setup depicted in Fig. 4 and record the data packet streams using wireshark at the receiver [9]. Motivated by the event counts in Fig. 9, we investigate next the bit rate of the network stream between the triangulation module and tracking receiver.

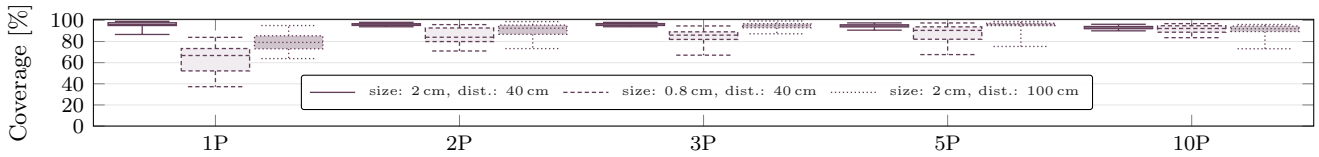
To characterize the burstiness of the traffic streams we use the concept of the *empirical traffic envelopes* [4]. Given a data traffic trace of one data flow with timestamped data packets where at time t_i a packet of size l_i is observed. The empirical traffic envelope $E(\tau)$ over a time window $\tau > 0$ is defined as

$$E(\tau) = \max_{t \in [t_{\mathrm{start}}, t_{\mathrm{end}})} \sum_{i: t_i \in [t, t+\tau)} l_i \tag{8}$$

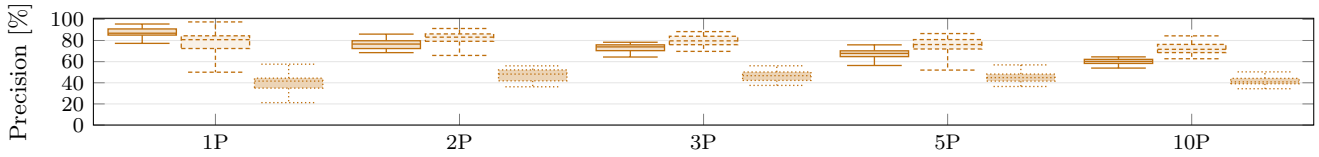
with t denoting any time point between the beginning of the trace t_{start} and the last timestamp in the trace t_{end}. In a nutshell, for every time window length τ, the envelope $E(\tau)$ returns the maximum number of bits observed over all possible windows of given length τ in the trace. In addition, $E(\tau)/\tau$ directly gives the *maximum bit rate observed in any window of length* τ. Note that the value $E(\tau^*)/\tau^*$ with τ^* being the entire length of the trace is the *average bit rate of the stream.*

Results: First, we report the average bit rates for different numbers of particles as {82 kbps, 268 kbps, 338 kbps, 310 kbps, 845 kbps} for {1, 2, 3, 5, 10}P, respectively. One may assume that the general trend in terms of the average bit rate increase is approximately linear in the number of particles as the events generated by the different packets are mostly independent. We note, however, that the events transmitted are the 3D reconstructed ones, hence, noise in the 3D reconstruction impacts the number of generated events and hence the uniformity of the packet contents. Interestingly, the triangulation matching at the sender side leads to the observed low data rates as it significantly filters the raw events as seen in Fig. 9.

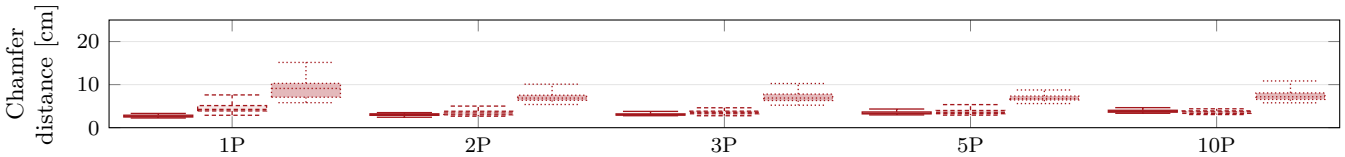
We investigate the burstiness of the event stream by computing the empirical traffic envelope $E(\tau)$ from (8) and plotting the corresponding maximum bit rate $E(\tau)/\tau$ over increasing time scales τ in Fig. 10. As a baseline, note that a constant bit rate flow would show $E(\tau)/\tau$ as a flat horizontal line in Fig. 10. Essentially, the difference



(a) Coverage of the trajectory points: particle size 2 cm, distance 40 cm; particle size 0.8 cm, distance 40 cm; particle size 2 cm, distance 100 cm.

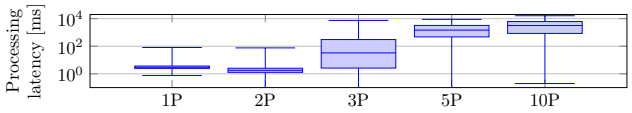


(b) Precision of the trajectory points: particle size 2 cm, distance 40 cm; particle size 0.8 cm, distance 40 cm; particle size 2 cm, distance 100 cm.

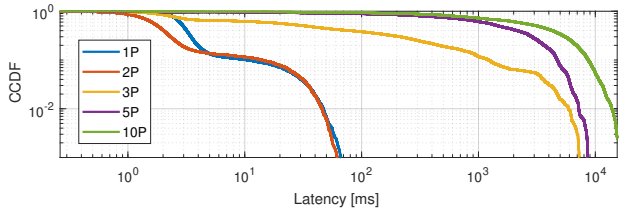


(c) Chamfer distance of the trajectory points: particle size particle size 2 cm, distance 40 cm; particle size 0.8 cm, distance 40 cm; particle size 2 cm, distance 100 cm.

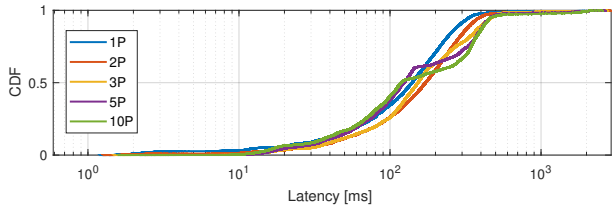
Figure 7: Accuracy evaluation with respect to the trajectory tracking output points for different particle counts $\{1, 2, 3, 5, 10\}$P. Comparison of coverage, precision and Chamfer distance over particle counts for trajectory point clouds, for 2 cm and 8 mm particle size and 40 cm and 100 cm distance between particle box and sensors.



(a) Box plots of the processing latency per particle count.



(b) CCDF of the processing latency per particle count.



(c) CDF of the remote tracking latency.

Figure 8: Empirical evaluation of the measured processing and remote tracking latencies.

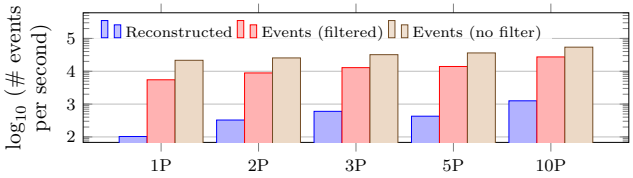


Figure 9: Average number of events per second for the different particle counts for one camera, i.e., *(i)* without DVS noise filtering, *(ii)* with DVS noise filtering, and *(iii)* the number of reconstructed points.

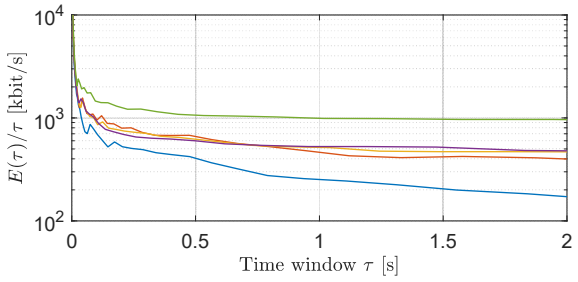


Figure 10: Burstiness evaluation for the event streams: The envelope rate $E(\tau)/\tau$ describes the burstiness, i.e., the maximum bit rate over any time window of length τ in the network stream. For a constant bitrate flow $E(\tau)/\tau$ is a flat line (no burstiness).

between the highest and lowest rate over different time scales τ *characterizes the flow burstiness* which intuitively translates to comparing the peak rate and the average rate of the flow when only taking the smallest and the largest τ in Fig. 10. We find that this difference spans *orders of magnitude* for 1P and is obviously reduced for 10P, i.e. the traffic flow becomes smoother, with more particles, as seen in Fig. 10. This burstiness translates to significantly different bit rates of arriving events at the tracking algorithm over time may explain that the algorithm sporadically resets the trajectories, e.g. when input events are missing for some time span.

6.4 Subsampling & Congestion Event Filtering

Subsampling the Event Streams: Previous work on streaming raw events from dynamic vision sensors has shown that remote,

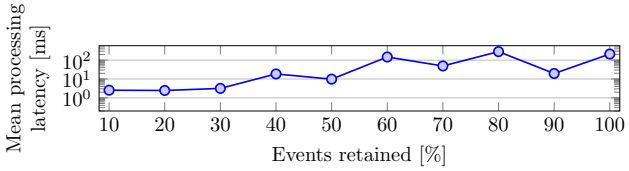


Figure 11: Random uniform subsampling a fraction κ of the raw events to be retained while discarding the rest of the raw events before transmission at the cameras (for 10P). We observe a significant decrease in processing (triangulation) latency with decreasing κ.

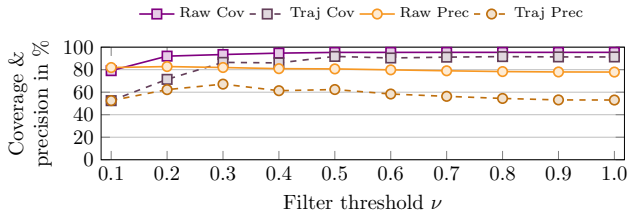
automated tasks, such as computer vision applications, retain functionality when receiving only a fraction of the raw events [10]. In this experiment we analyze the impact of *random event subsampling* at the sensors on the particle trajectory tracking as well as the processing latency. We conduct experiments with a 10P setup for 10 s where only a fraction $\kappa \in (0, 1]$ of the raw events are retained for triangulation, i.e., events are retained uniformly at random with probability κ per raw event per sensor.

The coverage and precision of the raw points are consistently high even for a small fraction κ of retained points while the trajectory points achieved lower coverage values at decreasing κ as ground truth points find less reconstructed points. As the precision remains unchanged across κ, the trajectory points still reliably match the ground truth points, indicating that the tracking quality is preserved. In Fig. 11 we show that the mean of the processing, i.e. triangulation, latency is reduced by *orders of magnitude* for decreasing κ. In a nutshell, we find that the remote particle tracking application is robust with respect to raw event uniform random subsampling that significantly accelerates processing.

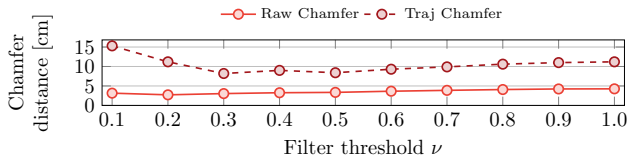
Score-based Filtering: In contrast to uniform subsampling of the raw events, we explore next, a different event filtering technique which utilizes the quality score $s \in [0, 1]$ from (2) of every triangulated event (matched events in both cameras) which is minimized for the best match. A higher score s indicates a higher probability of the triangulated event being produced by noise. The quality score filter discards events with a score higher than a threshold ν, i.e., $s \geq \nu$, which reduces the number of low-quality or noisy matches, and hence, the amount of transmitted events and increases the reconstruction accuracy.

Fig. 12 shows the mean of the accuracy metrics when varying the filter thresholds for experiments with 1 particle. The coverage and the precision for the raw points and trajectory points are stable after a steep increase for the first steps of the filter threshold ν. The Chamfer distance of the raw points remains agnostic to filtering, likely being caused by the density of the raw point cloud, while it shows some inaccuracies for the trajectory points, likely from imprecise predictions of the tracker due to sparse data.

Network Congestion-based Event Filtering: Finally, we use the congestion signal provided by the QUIC transport protocol in form of *pacing information*, i.e. the current *estimate by the congestion control algorithm* of a maximum burstiness transmission rate, to instruct the event-matching pipeline to adapt the quality score-based event filter. The rationale behind this feedback is to adapt the *event production rate out of the application to the congestion*



(a) Coverage and precision of raw and trajectory points.



(b) Chamfer distance of raw and trajectory points.

Figure 12: Mean of the accuracy metrics retrieved by different subsampling based on quality score thresholds, obtained by 25 experiments each for 10 seconds and 1P.

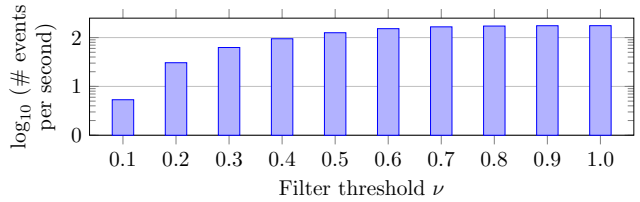


Figure 13: Average number of reconstructed events/sec for different filter thresholds ν given 1P.

state of the network. Essentially, if the network is not congested we transmit all triangulated events with a quality score $s \leq 0.5$ as we observe an almost flattening of the performance beyond this value in Fig. 12. If congestion is detected we tighten the filter threshold in this example to $s \leq 0.25$ or further to $s \leq 0.1$ to only transmit the triangulated events with the highest probability of being true, i.e., not caused by noise. We set the congestion control in TQUIC [33] to one of the following algorithms {CUBIC, BBR3, COPA} and read out the path statistics [32], which expose the pacing rate.

In the example in Fig. 14, we create artificial congestion on the network path using `tc` in four consecutive phases of 20 seconds each, going from 5 Mbps to 500 kbps to 100 kbps and 5 Mbps again. We choose these rates in accordance with the maximum envelope rates from Sect. 6.3 to induce congestion. Here, we couple the pacing rate to the score based filter such that when the pacing rate is less than 2 Mbps or 4 Mbps the filter threshold ν in the triangulation application is set more restrictively to 0.1 or 0.25, repectively. The trade-off between the filter threshold and the accuracy is shown in Fig. 12 while the average number of created triangulated events per second is given in Fig. 13. Fig. 14 shows the congestion aware filter threshold ν and the corresponding application output rates measured as the number of triangulated event points per second during the congestion periods for different congestion control algorithms. We observe that the number of events per second roughly follows the congestion signal, i.e., the pacing rate, even during probing phases of the congestion control algorithm. This network probing

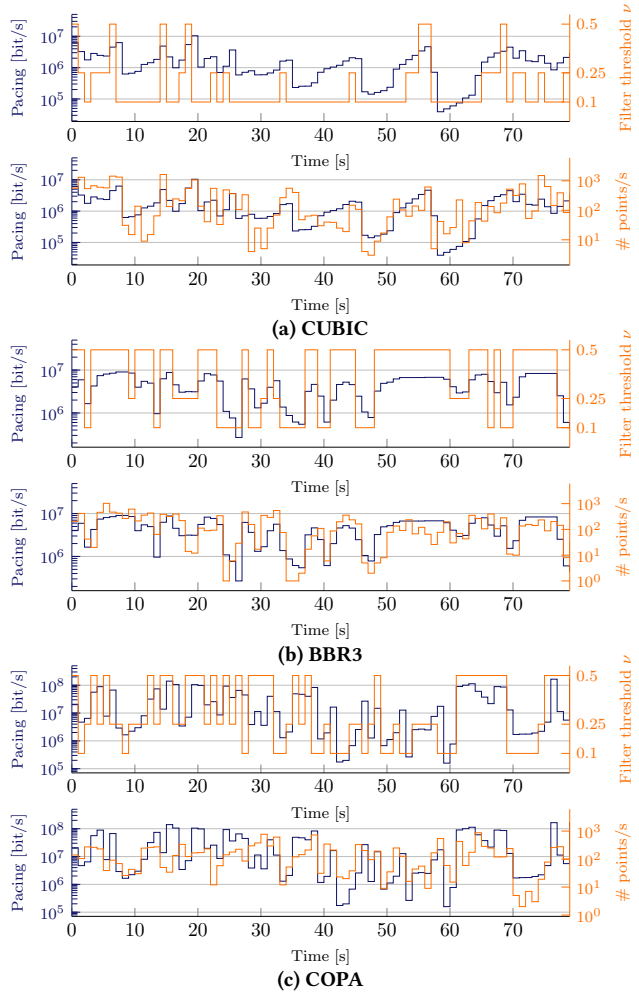


Figure 14: Pacing rate of different congestion control algorithms is used as feedback signal to adapt the quality score-based filtering of triangulated events, i.e., given congestion we only send high quality events (with score below ν). Observe that the pacing contains bandwidth probing segments that affect the filtering score. The available bandwidth is reduced in 20s steps with an increase in the last 20s. This experiment uses 2P.

in turn affects the quality score threshold leading to sending more noisy events to inject packets for available bandwidth estimation.

7 Limitations

In the following we discuss some of the limitations of this study. First, the limitations of the coupling between the event quality score and the congestion signal include a (currently) hard-wired mapping of pacing rate to quality score. In future this mapping may be improved through an event quality model. This may relate the contribution of the events considered to filtering to the quality of a downstream task, for example a computer vision task. This leads to a semantic communication approach where the events that contribute the most to the downstream task are transmitted under network congestion constraints.

Another limitation of this work is that deploying the system to different domains requires automatically recalibrating/adapting the parameters, such as the clustering radius and disparity limits, to the scene and particle size which may be automated. Furthermore, neural optimization of the weights for the event matching score could be investigated. Finally, the setup could be extended with further physical configurations, such as higher particle counts, or adjustable air flow patterns, to prepare for deployment in different use-cases.

8 Conclusion

This work is motivated by the rise of event-based vision sensors as tailored low-latency, high dynamic range capturing systems for machine-vision applications as an alternative to frame-based vision approaches. We presented a remote particle tracking system based on streams originating from stereo event-based vision sensors. The system reconstructs 3D particle positions from the 2D-temporal pixel event streams per sensor with an event matching algorithm and triangulation. Further, the system streams the triangulated points over the network to a remote particle tracking algorithm that takes filtered and processed events to estimate particle trajectories.

We investigate performance metrics of the system such as the tracking accuracy, streaming latency and throughput. We consider the accuracy of the raw reconstructed events as well as of the tracked trajectories against the ground truth obtained from an infrared motion capture system. The evaluation showed that the system can achieve high reconstruction accuracy while we observe that the tracking accuracy reduces with the number of particles.

We analyze the processing latency in the time synchronized system revealing a bottleneck in the event matching with an increasing number of events. Streaming and tracking latency analysis showed that most 3D points are remote tracked at the receiver within a latency of ~ 300 ms with a slight shift in quantiles with increasing number of particles. Further, the evaluation of the streaming throughput shows that the event streams are highly bursty and that this burstiness is *smoothed-out* with increasing number of particles. We attribute sporadic resets of trajectories by the tracking algorithm to this burstiness.

Random uniform subsampling of the raw event streams before triangulation significantly reduces the triangulation and processing latency without sacrificing tracking accuracy as given by the Chamfer distance between the tracked trajectories and the ground truth. Further, we couple the network congestion signal obtained from the QUIC transport layer to an adaptive event filtering to reduce the number of transmitted triangulated events during network congestion while still transmitting the events that are least likely to be corrupted by noise.

Acknowledgments

AI tools were used as an assistive aid during implementation and code refactoring. All method design, evaluations, and interpretations were conducted by the authors. The authors take full responsibility for the final content of this paper.

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